Quadratic Function Report

Polynomial

$$f(q) = aq^2 + bq + c$$

Naive

$$g_{\text{naive}}(X) = X^2a + Xb + c$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 2a\frac{\Delta}{\epsilon}^2 + aq^2 + bq + c$$

$$\mathrm{Var}(g_{\text{naive}}(X)) = 2\frac{\Delta}{\epsilon}^2 \left(10a^2\frac{\Delta}{\epsilon}^2 + 4a^2q^2 + 4abq + b^2\right)$$

$$\mathrm{MSE}(g_{\text{naive}}) = 2\frac{\Delta}{\epsilon}^2 \left(12a^2\frac{\Delta}{\epsilon}^2 + 4a^2q^2 + 4abq + b^2\right)$$

$$\mathrm{Bias}(g_{\text{naive}}) = 2a\frac{\Delta}{\epsilon}^2$$

Unbiased

$$g_{\text{unb}}(X) = X^2a + Xb - 2a\frac{\Delta}{\epsilon}^2 + c$$

$$\mathbb{E}[g_{\text{unb}}(X)] = aq^2 + bq + c$$

$$\mathrm{Var}(g_{\text{unb}}(X)) = 2\frac{\Delta}{\epsilon}^2 \left(10a^2\frac{\Delta}{\epsilon}^2 + 4a^2q^2 + 4abq + b^2\right)$$

$$\mathrm{MSE}(g_{\text{unb}}) = 2\frac{\Delta}{\epsilon}^2 \left(10a^2\frac{\Delta}{\epsilon}^2 + 4a^2q^2 + 4abq + b^2\right)$$

Comparison

$$\text{mean gap} = -2a\frac{\Delta}{\epsilon}^{2}$$

$$\text{variance gap} = 0$$

$$\text{MSE gap} = -4a^2\frac{\Delta}{\epsilon}^{4}$$

$$\text{Bias}(g_{\text{naive}})^2 = 4a^2\frac{\Delta}{\epsilon}^{4}$$